

Лабораторная работа №1

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```
RED OS 8.0.2
Kernel 6.12.21-1.red00.x86_64 on an x86_64

localhost login: redos
25Password:
Last login: Fri Sep 12 15:10:17 on tty2
[redos@localhost ~]$ whoami
redos
[redos@localhost ~]$ whoain
-bash: whoain: command not found
[redos@localhost ~]$ whoani
redos
[redos@localhost ~]$ echo "Hello world"
Hello world
[redos@localhost ~]$ date
Fri Sep 12 03:25:53 PM MSK 2025
[redos@localhost ~]$ uptime
 15:26:00 up 47 min,  3 users,  load average: 0.12, 0.29, 0.52
[redos@localhost ~]$
```

Выполнение пунктов 1.1-1.9

```

UPTIME(1)                                User Commands                                UPTIME(1)

NAME
    uptime - Tell how long the system has been running.

SYNOPSIS
    uptime [options]

DESCRIPTION
    uptime gives a one line display of the following information. The current time, how long the system has been running, how many users are currently
    logged on, and the system load averages for the past 1, 5, and 15 minutes.

    This is the same information contained in the header line displayed by w(1).

    System load averages is the average number of processes that are either in a runnable or uninterruptable state. A process in a runnable state is ei
    ther using the CPU or waiting to use the CPU. A process in uninterruptable state is waiting for some I/O access, eg waiting for disk. The averages
    are taken over the three time intervals. Load averages are not normalized for the number of CPUs in a system, so a load average of 1 means a single
    CPU system is loaded all the time while on a 4 CPU system it means it was idle 75% of the time.

OPTIONS
    -p, --pretty
        show uptime in pretty format

    -h, --help
        display this help text

    -s, --since
        system up since, in yyyy-mm-dd HH:MM:SS format

    -V, --version
        display version information and exit

FILES
    /var/run/utmp
        information about who is currently logged on

    /proc
        process information

AUTHORS
    uptime was written by Larry Greenfield <greenfie@gauss.rutgers.edu> and Michael K. Johnson <johnsonm@sunsite.unc.edu>

SEE ALSO
    ps(1), top(1), utmp(5), w(1)

REPORTING BUGS
    Please send bug reports to <procps@freelists.org>

procps-ng                                December 2012                                UPTIME(1)

```

man uptime

```

(redos@localhost ~)$ history
1  whoami
2  whoami
3  whoami
4  echo "Hello World"
5  echo "Hello World"
6  date
7  uptime
8  sudo systemctl isolate multi-user.target
9  whoami
10 whoami
11 whoami
12 echo "Hello World"
13 date
14 uptime
15 u
16 uptime
17 sudo systemctl isolate graphical.target
18 uptime
19 u
20 sudo systemctl isolate multi-user.target
21 whoami
22 whoami
23 whoami
24 echo "Hello world"
25 date
26 uptime
27 man uptime
28 q
29 wdate
30 uptime
31 q
32 clear
33 history
(redos@localhost ~)$

```

```

(redos@localhost ~)$ echo 123 > file
(redos@localhost ~)$ cat file
123
(redos@localhost ~)$ echo 456 >> file
(redos@localhost ~)$ cat file
123
456
(redos@localhost ~)$ _

```

1.13

ЗАДАНИЕ 2

```

(redos@localhost ~)$ pwd
/home/redos
(redos@localhost ~)$ cd /
(redos@localhost ~)$ cd ..
(redos@localhost ~)$ pwd
/
(redos@localhost ~)$ ls
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin srv sys tmp usr var
(redos@localhost ~)$ ls -a
. .. bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin srv sys tmp usr var
(redos@localhost ~)$ ls /var
account adm cache db empty ftp games kerberos lib local lock log mail nis opt preserve run spool target tmp yp
(redos@localhost ~)$ cd /
(redos@localhost ~)$ pwd
/home
(redos@localhost ~)$ ls
redos
(redos@localhost ~)$ ls -a
. .. redos
(redos@localhost ~)$ _

```

Выполнение заданий 2.1-2.5

```

[redos@localhost home]$ cd ~
[redos@localhost ~]$ mkdir fruits
[redos@localhost ~]$ cd /
[redos@localhost /]$ ls
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin srv sys tmp usr var
[redos@localhost /]$ mkdir ~/animals
[redos@localhost /]$ touch tmp/temp
[redos@localhost /]$ cd ~/fruits
[redos@localhost fruits]$ touch apple banana pineapple lion
[redos@localhost fruits]$ touch ../animals/cat.txt ../animals/dog.txt ../animals/elephant.txt
[redos@localhost fruits]$ ls -l apple
-rw-r--r--. 1 redos redos 0 Sep 12 16:33 apple
[redos@localhost fruits]$ ls b*
banana
[redos@localhost fruits]$ ls *a
banana
[redos@localhost fruits]$ ls *[bi]*
banana lion pineapple
[redos@localhost fruits]$ ls b* i*
ls: cannot access 'i*': No such file or directory
banana
[redos@localhost fruits]$ cp /etc/passwd ~
[redos@localhost fruits]$ cat /etc/issue
NS
Kernel \r on an \m
[redos@localhost fruits]$ cp /etc/issue ./apple
[redos@localhost fruits]$ cat apple
NS
Kernel \r on an \m
[redos@localhost fruits]$ mv lion ../animals/

```

Выполнение 2.7 - 3.16

```

[redos@localhost fruits]$ mv pineapple pineapple
[redos@localhost fruits]$ wc -l /etc/passwd
50 /etc/passwd
[redos@localhost fruits]$ _

```

Выполнение 3.17-3.18

```
(redos@localhost fruits)$ cd ~
(redos@localhost ~)$ ls -l passwd_link
lrwxrwxrwx. 1 redos redos 11 Sep 12 17:06 passwd_link -> /etc/passwd
(redos@localhost ~)$ ls -l .bash_history history_hard
-rw-r-----. 2 redos redos 264 Sep 12 15:23 .bash_history
-rw-r-----. 2 redos redos 264 Sep 12 15:23 history_hard
(redos@localhost ~)$
```

3.19-3.21

```
(redos@localhost ~)$ rm fruits/apple
(redos@localhost ~)$ rmdir rme
(redos@localhost ~)$ rmdir rme
(redos@localhost ~)$ rmdir fruits
rmdir: failed to remove 'fruits': Directory not empty
(redos@localhost ~)$ rm -fr fruits
(redos@localhost ~)$ _
```

3.22-3.25

```

daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System Message Bus:/usr/sbin/nologin
rtkit:x:172:172:RealtimeKit:/proc:/sbin/nologin
geoclue:x:999:999:User for geoclue:/var/lib/geoclue:/sbin/nologin
pipewire:x:998:998:PipeWire System Daemon:/run/pipewire:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
systemd-oom:x:997:997:systemd Userspace OOM Killer:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
unbound:x:996:995:Unbound DNS resolver:/etc/unbound:/sbin/nologin
sstpc:x:995:994:Secure Socket Tunneling Protocol (SSTP) Client:/var/run/sstpc:/sbin/nologin
colord:x:994:993:User for colord:/var/lib/colord:/sbin/nologin
postfix:x:89:89:/var/spool/postfix:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/usr/sbin/nologin
nm-openconnect:x:993:992:NetworkManager user for OpenConnect:/:/sbin/nologin
openswitch:x:992:991:Open uSwitch Daemons:/:/sbin/nologin
gluster:x:991:989:GlusterFS daemons:/run/gluster:/sbin/nologin
dnsmasq:x:990:987:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/sbin/nologin
saslauthd:x:989:76:Saslauthd user:/run/saslauthd:/sbin/nologin
openvpn:x:988:984:OpenVPN:/etc/openvpn:/sbin/nologin
nm-openvpn:x:987:983:Default user for running openvpn spawned by NetworkManager:/:/sbin/nologin
abrt:x:173:173:/etc/abrt:/sbin/nologin
usdd:x:986:982:Web Services Dynamic Discovery host daemon:/:/sbin/nologin
gdm:x:42:42:GNOME Display Manager:/var/lib/gdm:/usr/sbin/nologin
firebird:x:985:981:/usr/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
chromy:x:984:979:/var/lib/chromy:/sbin/nologin
tcpdump:x:72:72:tcpdump:/usr/sbin/nologin
gnome-remote-desktop:x:975:975:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
systemd-coredump:x:974:974:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:973:973:systemd Time Synchronization:/usr/sbin/nologin
uboxadd:x:972:1:/var/run/uboxadd:/usr/sbin/nologin
redos:x:1000:1000:redos:/home/redos:/bin/bash
nslcd:x:65:55:LDAP Client User:/:/sbin/nologin
passin:x:971:971:Local Caching Server:/usr/share/empty:/usr/sbin/nologin

```

3.26

```

[redos@localhost ~]$ cp -r /etc/init.d/ ~
[redos@localhost ~]$ ls -ld init.d
drwxr-xr-x. 2 redos redos 37 Sep 12 17:34 init.d
[redos@localhost ~]$ rm -fr init.d
[redos@localhost ~]$ ls
animals Desktop Documents Downloads file history_hard Music passwd passwd_link Pictures Public Templates Videos
[redos@localhost ~]$ _

```

3.27-3.28

```
(redos@localhost ~)$ cut -d: -f1 /etc/passwd | sort_
```



```
adm
avahi
bin
chrony
colord
daemon
dbus
dnsmasq
firebird
ftp
games
gdm
geoclue
gluster
gnome-remote-desktop
halt
lp
mail
nm-openconnect
nm-openvpn
nobody
nslcd
openvpn
openvswitch
operator
passim
pipewire
polkitd
postfix
redos
root
rpc
rpcuser
rtkit
saslauthd
shutdown
sshd
ssstpc
sync
systemd-coredump
systemd-oom
systemd-resolve
systemd-timesync
tcpdump
tss
unbound
usbmuxd
vboxadd
```

4.1

```
(redos@localhost ~)$ cut -d: -f1,3 /etc/passwd | sort_
```

```
adm:3  
avahi:78  
bin:1  
chrony:984  
colord:994  
daemon:2  
dbus:81  
dnsmasq:998  
firebird:985  
ftp:14  
games:12  
gdm:42  
geoclue:999  
guster:991  
gnome-remote-desktop:975  
halt:7  
lp:4  
mail:8  
nm-openconnect:993  
nm-openvpn:987  
nobody:65534  
nslcd:65  
openvpn:988  
openswitch:992  
operator:11  
passim:971  
pipewire:998  
polkitd:114  
postfix:89  
redos:1000  
root:0  
rpc:32  
rpcuser:23  
rtkit:172  
saslauth:989  
shutdown:6  
sshd:74  
sspc:995  
sync:5  
systemd-coredump:974  
systemd-oom:997  
systemd-resolve:193  
systemd-timesync:973  
tcpdump:72  
tss:59  
unbound:996  
usbmuxd:113  
vboxadd:972
```

4.2

```
(redos@localhost ~)$ grep :/bin/bash /etc/passwd | wc -l
2
(redos@localhost ~)$
```

4.3

```
(redos@localhost ~)$ grep "/bin/bash" /etc/passwd | cut -d: -f1 | sort -r
root
redos
(redos@localhost ~)$
```

4.4

```
fredos@localhost ~]$ sort /etc/passwd | tr ":" " " _
```

```
adm x 3 4 adm /var/adm /sbin/nologin
avahi x 70 70 Avahi mDNS/DNS-SD Stack /var/run/avahi-daemon /sbin/nologin
bin x 1 1 bin /bin /sbin/nologin
chrony x 984 979 /var/lib/chrony /sbin/nologin
colord x 994 993 User for colord /var/lib/colord /sbin/nologin
daemon x 2 2 daemon /sbin /sbin/nologin
dbus x 81 81 System Message Bus /usr/sbin/nologin
dnsmasq x 999 997 Dnsmasq DHCP and DNS server /var/lib/dnsmasq /usr/sbin/nologin
firebird x 985 981 /usr/sbin/nologin
ftp x 14 50 FTP User /var/ftp /sbin/nologin
games x 12 100 games /usr/games /sbin/nologin
gdm x 42 42 GNOME Display Manager /var/lib/gdm /usr/sbin/nologin
geoclue x 999 999 User for geoclue /var/lib/geoclue /sbin/nologin
gluster x 991 989 GlusterFS daemons /run/gluster /sbin/nologin
gnome-remote-desktop x 975 975 GNOME Remote Desktop /var/lib/gnome-remote-desktop /usr/sbin/nologin
halt x 7 0 halt /sbin /sbin/halt
lp x 4 7 lp /var/spool/lpd /sbin/nologin
mail x 0 12 mail /var/spool/mail /sbin/nologin
nm-openconnect x 993 992 NetworkManager user for OpenConnect / /sbin/nologin
nm-openvpn x 987 983 Default user for running openvpn spawned by NetworkManager / /sbin/nologin
nobody x 65534 65534 Kernel Overflow User / /sbin/nologin
nslcd x 65 55 LDAP Client User / /sbin/nologin
openvpn x 988 984 OpenVPN /etc/openvpn /sbin/nologin
openswitch x 992 991 Open vSwitch Daemons / /sbin/nologin
operator x 11 0 operator /root /sbin/nologin
passim x 971 971 Local Caching Server /usr/share/empty /usr/sbin/nologin
pipewire x 998 998 PipeWire System Daemon /run/pipewire /usr/sbin/nologin
polkitd x 114 114 User for polkitd / /sbin/nologin
postfix x 89 89 /var/spool/postfix /sbin/nologin
redos x 1000 1000 redos /home/redos /bin/bash
root x 0 0 root /root /bin/bash
rpcuser x 29 29 RPC Service User /var/lib/nfs /sbin/nologin
rpc x 32 32 Rpcbind Daemon /var/lib/rpcbind /sbin/nologin
rtkit x 172 172 RealtimeKit /proc /sbin/nologin
saslauthd x 989 76 Saslauthd user /run/saslauthd /sbin/nologin
shutdown x 6 0 shutdown /sbin /sbin/shutdown
sshd x 74 74 Privilege-separated SSH /usr/share/empty.sshd /usr/sbin/nologin
sstpc x 995 994 Secure Socket Tunneling Protocol(SSTP) Client /var/run/sstpc /sbin/nologin
sync x 5 0 sync /sbin /bin/sync
systemd-coredump x 974 974 systemd Core Dumper /usr/sbin/nologin
systemd-oom x 997 997 systemd Userspace OOM Killer /usr/sbin/nologin
systemd-resolve x 193 193 systemd Resolver /usr/sbin/nologin
systemd-timesync x 973 973 systemd Time Synchronization /usr/sbin/nologin
tcpdump x 72 72 tcpdump /usr/sbin/nologin
tss x 59 59 Account used for TPM access /usr/sbin/nologin
unbound x 996 995 Unbound DNS resolver /etc/unbound /sbin/nologin
usbmuxd x 113 113 usbmuxd user / /sbin/nologin
vboxadd x 972 1 /var/run/vboxadd /usr/sbin/nologin
```

4.6

```
fredos@localhost ~]$ cut -d: -f1,7 /etc/passwd | sort | tr " " " _
```

```
adm /sbin/nologin
avahi /sbin/nologin
bin /sbin/nologin
chrony /sbin/nologin
colord /sbin/nologin
daemon /sbin/nologin
dbus /usr/sbin/nologin
dnsmasq /usr/sbin/nologin
firebird /usr/sbin/nologin
ftp /sbin/nologin
games /sbin/nologin
gdm /usr/sbin/nologin
geoclue /sbin/nologin
gluster /sbin/nologin
gnome-remote-desktop /usr/sbin/nologin
halt /sbin/halt
lp /sbin/nologin
mail /sbin/nologin
nm-openconnect /sbin/nologin
nm-openvpn /sbin/nologin
nobody /sbin/nologin
nslcd /sbin/nologin
openvpn /sbin/nologin
openswitch /sbin/nologin
operator /sbin/nologin
passim /usr/sbin/nologin
pipewire /usr/sbin/nologin
polkitd /sbin/nologin
postfix /sbin/nologin
redos /bin/bash
root /bin/bash
rpc /sbin/nologin
rpcuser /sbin/nologin
rtkit /sbin/nologin
saslauthd /sbin/nologin
shutdown /sbin/shutdown
sshd /usr/sbin/nologin
ssnps /sbin/nologin
sync /bin/sync
systemd-coredump /usr/sbin/nologin
systemd-oom /usr/sbin/nologin
systemd-resolve /usr/sbin/nologin
systemd-timesync /usr/sbin/nologin
tcpdump /usr/sbin/nologin
tss /usr/sbin/nologin
unbound /sbin/nologin
usbmuxd /sbin/nologin
vboxadd /usr/sbin/nologin
```

```
(redos@localhost ~)$ ping ya.ru
PING ya.ru (5.255.255.242) 56(84) bytes of data.
64 bytes from ya.ru (5.255.255.242): icmp_seq=1 ttl=240 time=264 ms
64 bytes from ya.ru (5.255.255.242): icmp_seq=2 ttl=240 time=264 ms
64 bytes from ya.ru (5.255.255.242): icmp_seq=3 ttl=240 time=268 ms
^C
--- ya.ru ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2000ms
rtt min/avg/max/mdev = 263.867/265.317/267.750/1.730 ms
(redos@localhost ~)$ _
```

5.1

```
(redos@localhost ~)$ curl wttr.in
Weather report: not found

Overcast
13 °C
■ 7 km/h
10 km
0.0 mm



| Fri 12 Sep                                                   |                                                          |
|--------------------------------------------------------------|----------------------------------------------------------|
| Morning                                                      | Noon                                                     |
| Partly Cloudy<br>15 °C<br>■ 4-6 km/h<br>10 km<br>0.0 mm   0% | Sunny<br>22(24) °C<br>■ 4-5 km/h<br>10 km<br>0.0 mm   0% |
| Evening                                                      |                                                          |
| Cloudy<br>24(25) °C<br>■ 3 km/h<br>10 km<br>0.0 mm   0%      | Night                                                    |
| Clear<br>21 °C<br>■ 6-14 km/h<br>10 km<br>0.0 mm   0%        |                                                          |



| Sat 13 Sep                                            |                                                        |
|-------------------------------------------------------|--------------------------------------------------------|
| Morning                                               | Noon                                                   |
| Sunny<br>18 °C<br>■ 4-5 km/h<br>10 km<br>0.0 mm   0%  | Sunny<br>25 °C<br>■ 10-11 km/h<br>10 km<br>0.0 mm   0% |
| Evening                                               |                                                        |
| Sunny<br>23 °C<br>■ 8-9 km/h<br>10 km<br>0.0 mm   0%  | Night                                                  |
| Clear<br>20 °C<br>■ 7-14 km/h<br>10 km<br>0.0 mm   0% |                                                        |



| Sun 14 Sep                                                       |                                                                |
|------------------------------------------------------------------|----------------------------------------------------------------|
| Morning                                                          | Noon                                                           |
| Thundery outbr...<br>17 °C<br>■ 4-5 km/h<br>10 km<br>0.0 mm   0% | Thundery outbr...<br>20 °C<br>■ 2 km/h<br>10 km<br>0.0 mm   0% |
| Evening                                                          |                                                                |
| Cloudy<br>20 °C<br>■ 7-8 km/h<br>10 km<br>0.0 mm   0%            | Night                                                          |
| Partly Cloudy<br>19 °C<br>■ 9-20 km/h<br>10 km<br>0.0 mm   0%    |                                                                |



Follow @igor\_chubin for wttr.in updates
(redos@localhost ~)$
```

5.2

```
[redos@localhost ~]$ hostname -I
192.168.64.2 fd46:77fc:9ad8:f914:9ce7:4aff:fe15:df4f
[redos@localhost ~]$ curl ifconfig.me
77.229.194.169[redos@localhost ~]$ _
```

5.3

